

Automatic Fire Fighting & Rescue Robot

Abstract

This research paper presents the structural engineering, multisensory infrastructure, and autonomous mitigation logic of a high-efficiency hybrid Unmanned Ground Vehicle (UGV) and Airborne Unmanned Aerial Vehicle (UAV) "Automatic Fire Fighting and Rescue Robot" developed by Club Quantum Spark. The core architectural framework focuses on automated localized thermal detection, multi-agent fire suppression, advanced environmental gas sensing, and integrated human search-and-rescue routines. Fabricated with a high-temperature resistant carbon fiber chassis, the platform features a detachable autonomous drone for elevated fire suppression, a robotic manipulator arm for obstacle clearing, real-time telemetry control panels, and an audio-feed system for victim detection. The prototype successfully secured 2nd Place at the Sylhet Sadar Upazila Level and advanced to achieve 1st Place (Champion) at the District Level during the 47th National Science and Technology Week. It is now officially nominated for the international round at the World Invention Competition and Exhibition (WICE 2026) in Malaysia. This paper details the hardware configuration, fluid dynamics, multi-terrain locomotion, and spatial control loops that govern the platform's multi-functional emergency task execution.

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1. Introduction

Fire hazards in industrial zones, high-rise buildings, and dense residential sectors pose severe threats to human lives and structural property. Conventional rescue operations expose first responders to toxic gas inhalation, flashovers, and sudden structural collapses. To address these critical vulnerabilities, the field of robotics has shifted toward deploying Unmanned Ground Vehicles (UGVs) and Unmanned Aerial Vehicles (UAVs) to automate hazard mitigation.



To overcome the spatial limitations of standard ground robots, Club Quantum Spark has engineered an integrated, hybrid "Automatic Fire Fighting and Rescue Robot." The platform uniquely combines a heavy-duty ground unit with a detachable fire-suppression drone, enabling simultaneous ground and aerial intervention. Equipped with specialized sub-systems including a robotic arm for debris management, real-time gas analyzers, and live telemetry control, this platform establishes a scalable framework for modern disaster response.

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2. Mechanical Design and Multi-Terrain Locomotion Architecture

The structural chassis of the primary ground unit is fabricated using high-grade Carbon Fiber Composite Materials. This engineering choice guarantees superior thermal resistance against high-temperature radiation while minimizing structural inertia to maximize battery efficiency.



To navigate volatile, debris-ridden disaster zones, the locomotion system utilizes a dual-configuration terrain mechanism. It integrates a continuous Chain Track Wheel System for maximum traction over rubble, combined with high-velocity **Tube Wheels** for rapid transit across stable surfaces.

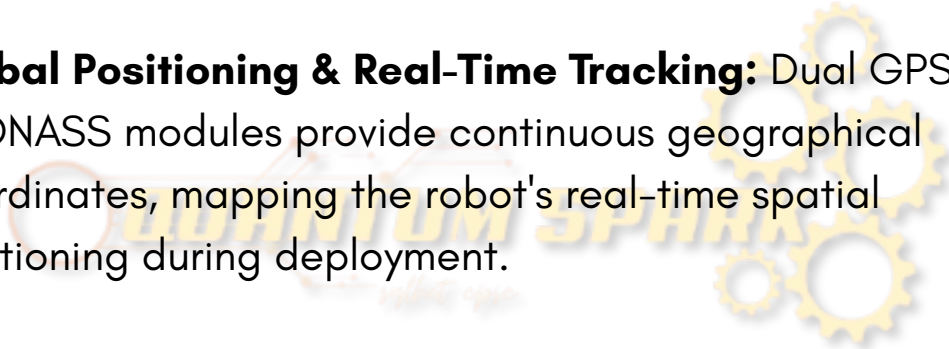
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3. Core Features and Technical Capabilities

The robotic platform operates through an advanced array of specific sub-systems designed for comprehensive hazard management:

- **Automated Fire Detection & Targeted Suppression:** The platform executes automated scanning for flame patterns. Upon positive triangulation, it self-navigates to the source and deploys specific fire-extinguishing agents tailored to the class of fire encountered.
- **Integrated Aerial UAV Sub-System (Detachable Drone):** The main chassis houses an automated drone. Upon remote or autonomous command, the drone executes a mechanical detach sequence, performs vertical takeoff, suppresses elevated fire vectors, and automatically returns to its designated landing deck on the ground robot.
- **Robotic Manipulator Arm:** A high-torque robotic arm is mounted on the front chassis to mechanically displace obstructions or clear debris blocking escape corridors.
- **Environmental Gas Analytics:** The system features an onboard gas sensor matrix that continuously analyzes ambient air quality, measuring hazardous gas compositions and transmitting real-time data directly to the central control panel.

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- **Humanitarian Audio-Rescue Feed:** An integrated microphone array and audio feedback loop allow rescue operators to monitor and detect high-frequency human distress vocalizations or screams within zero-visibility zones.
 - **Global Positioning & Real-Time Tracking:** Dual GPS-GLONASS modules provide continuous geographical coordinates, mapping the robot's real-time spatial positioning during deployment.
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4. Control Infrastructure and Live Telemetry Monitoring

The robot features dual-mode operability, utilizing an autonomous sensory loop alongside an extended-range Remote Control Framework. Live diagnostic telemetry, sensory feeds, gas density data, and navigation parameters are continuously streamed back to an advanced, user-friendly Control Panel Graphical User Interface (GUI).

To ensure precise path correction and steering response under extreme conditions, the orientation vectors are continuously adjusted using a tuned Proportional-Integral-Derivative (PID) loop:

$$U(t) = K_p \times e(t) + K_i \times \int e(t)dt + K_d \times \frac{de(t)}{dt}$$

(Where $e(t)$ is the displacement error from the optimal safe trajectory, and K_p , K_i , K_d represent the tuned control coefficients).

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5. Milestones and Project Evaluation

The experimental prototype has undergone rigorous technical evaluations across multiple competitive institutional phases:

- Upazila Level: Achieved 2nd Place at the Sylhet Sadar Upazila regional round within the 47th National Science and Technology Week, demonstrating outstanding computational logic and core technical design among competing institutions.
- District Level: Successfully advanced to win 1st Place (Champion) at the District Level, receiving high technical commendation from the central evaluation panels for its integrated hybrid drone-ground configuration.
- International Qualification: Formally nominated and selected by the national board to represent Bangladesh at the upcoming World Invention Competition and Exhibition (WICE 2026) in Malaysia.

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6. Conclusion and Future Work

This research presents a highly reliable, hybrid robotics solution for automated firefighting and civilian rescue. By integrating a carbon fiber ground chassis, a high-pressure fluid pump, a multi-sensor array, and a detachable autonomous drone, Club Quantum Spark has successfully developed a robust platform for modern emergency response. Future works will focus on integrating edge-AI computing matrices for autonomous human detection through thick smoke via thermal vision processing.



References :

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3. Technical Criteria and Innovation Standards Matrix, WICE 2026, Malaysia.

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